Introduction to Data Science

CSE 451
Advanced Topics in Computer Science
Fall 2023
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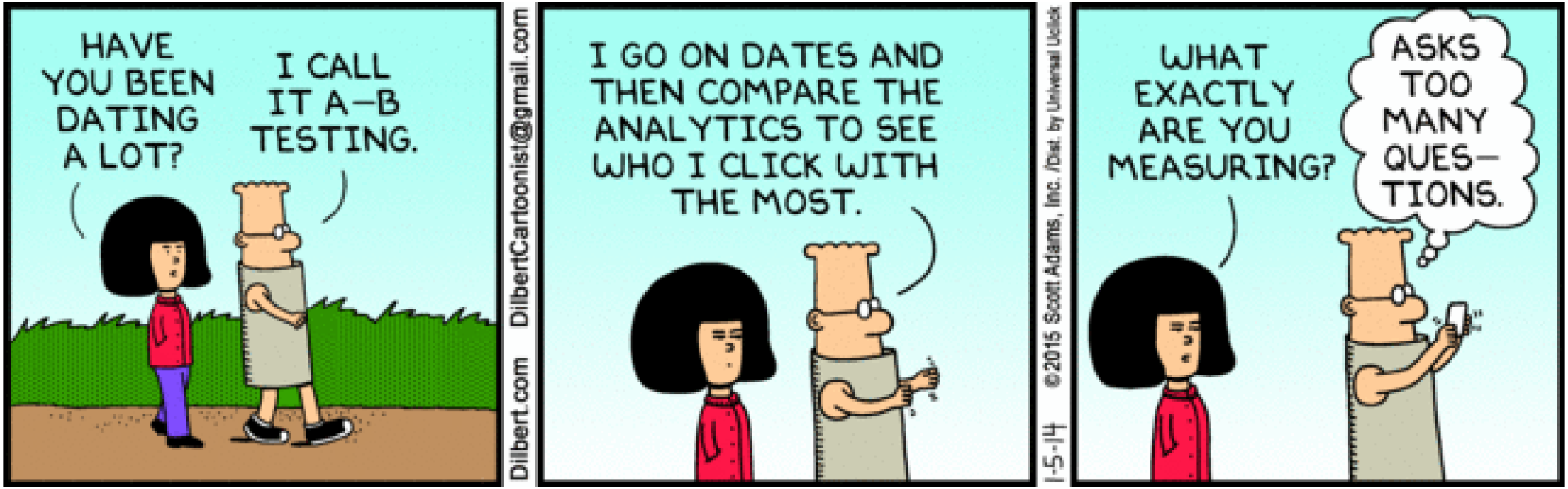
What do companies in e-commerce, entertainment, healthcare, manufacturing, marketing, finance, tech, and hundreds of other sectors (including government entities) all have in common?

What do companies in e-commerce, entertainment, healthcare, manufacturing, marketing, finance, tech, and hundreds of other sectors (including government entities) all have in common?

They all use data

Data is everywhere.
You <u>use</u> and <u>create</u> data everyday.



Have you ever read reviews of a product before deciding whether or not to buy it?

You are analyzing data.

Or maybe you wear a fitness tracker to count your steps so you can stay active throughout the day.

You are analyzing data.

But you don't just use data.

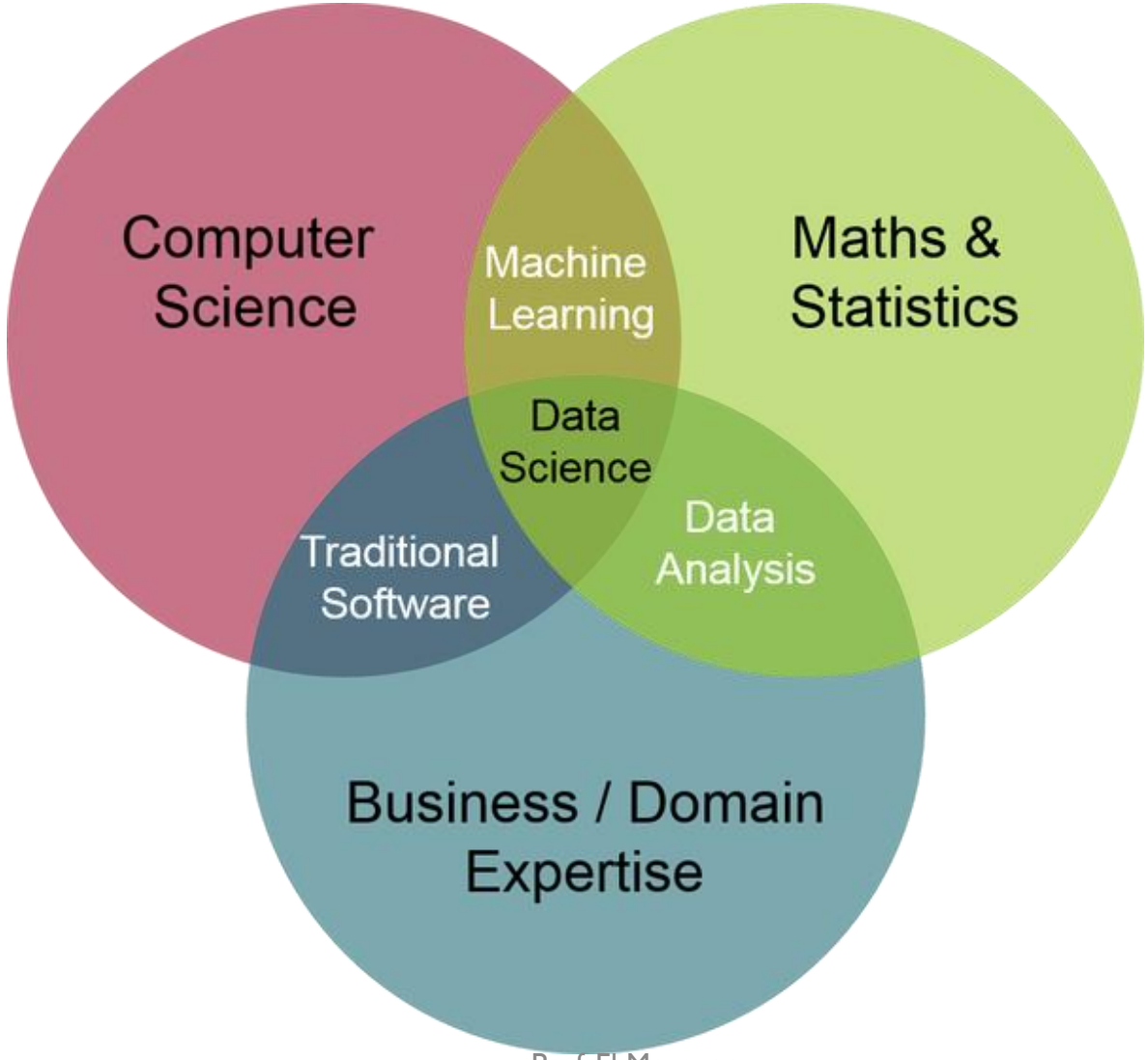
You also create <u>huge amounts</u> of it every single day.

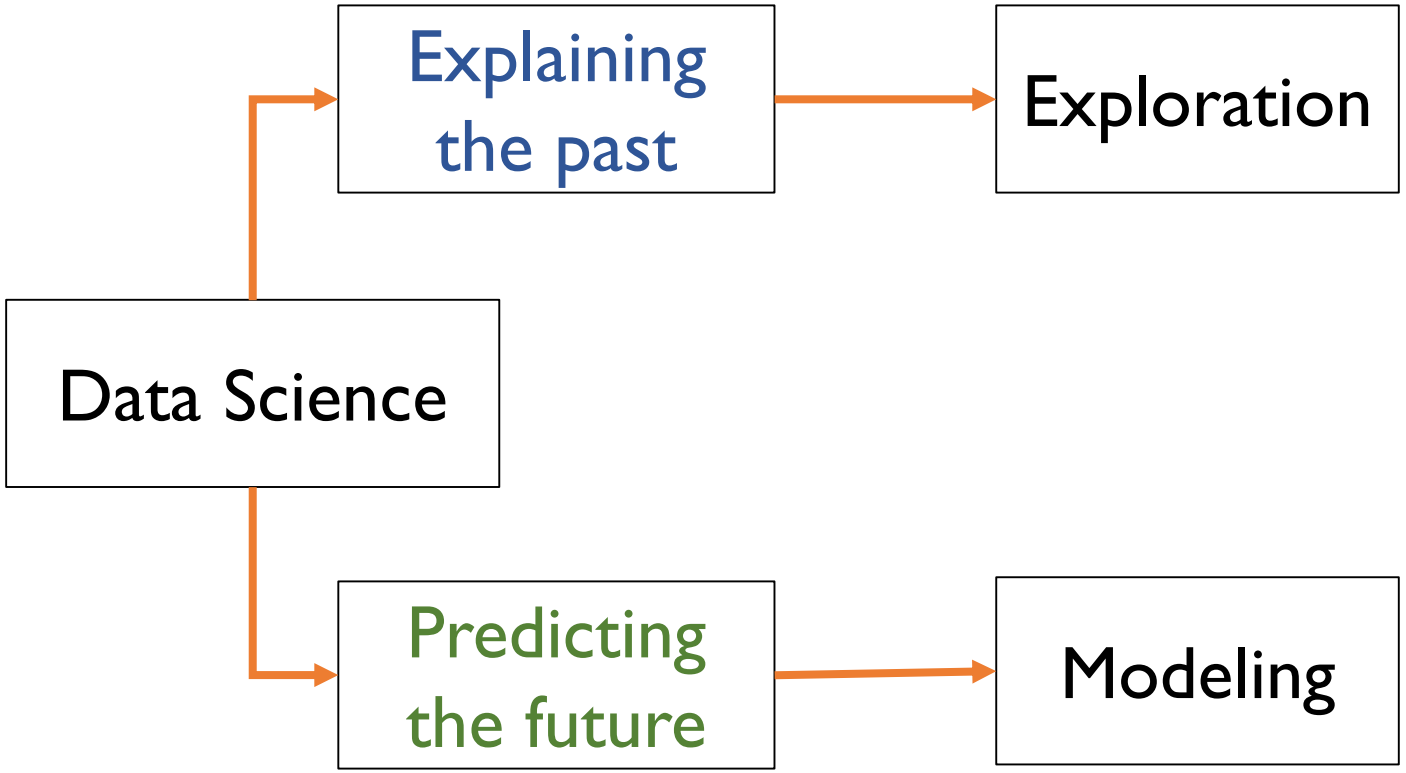
Any time you use your phone,
look up something online,
stream music,
shop with a credit card,
post on social media or
use GPS to map a route,

you're creating data.

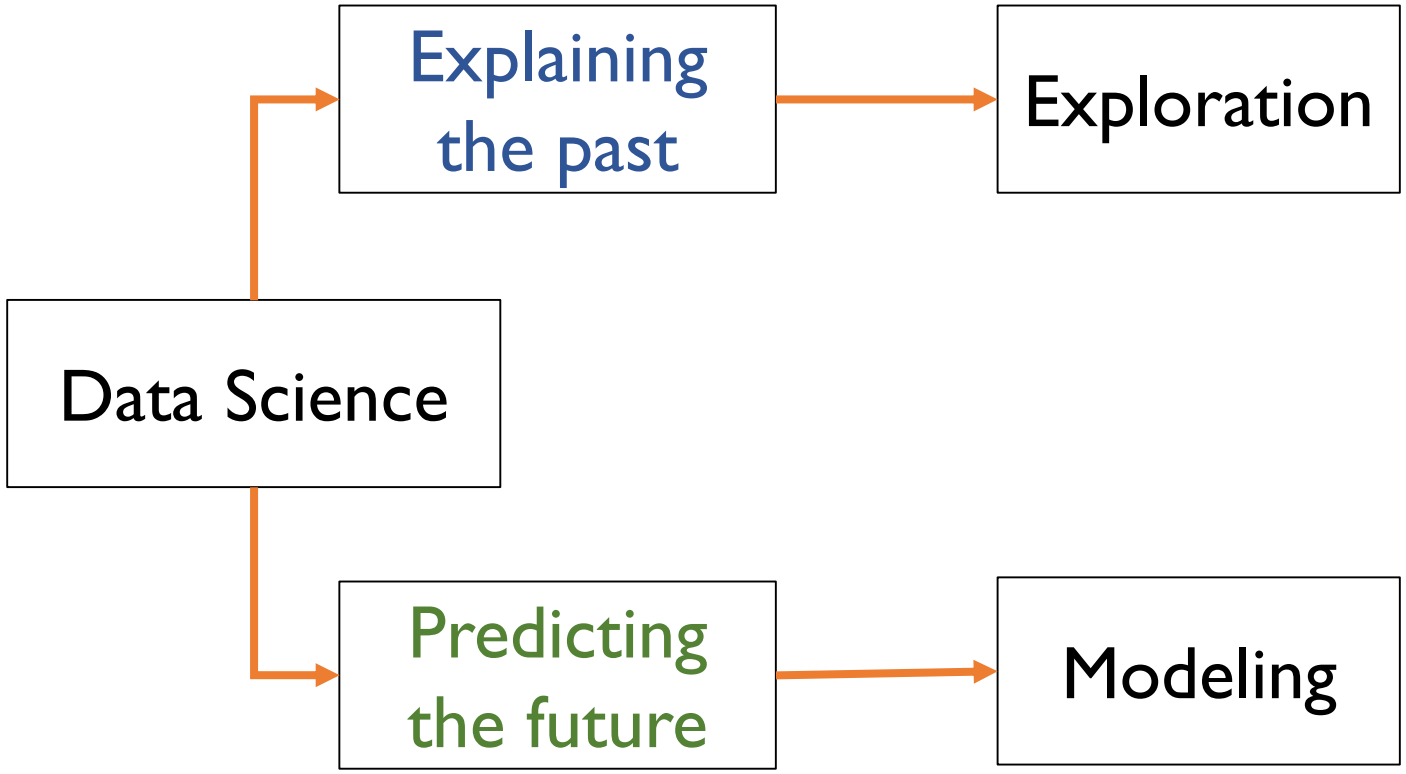
Q. What to do with all this data?

A. Dedicate a discipline to the study of data




Data Science
Explaining the past
Exploration
Predicting the future
Modeling

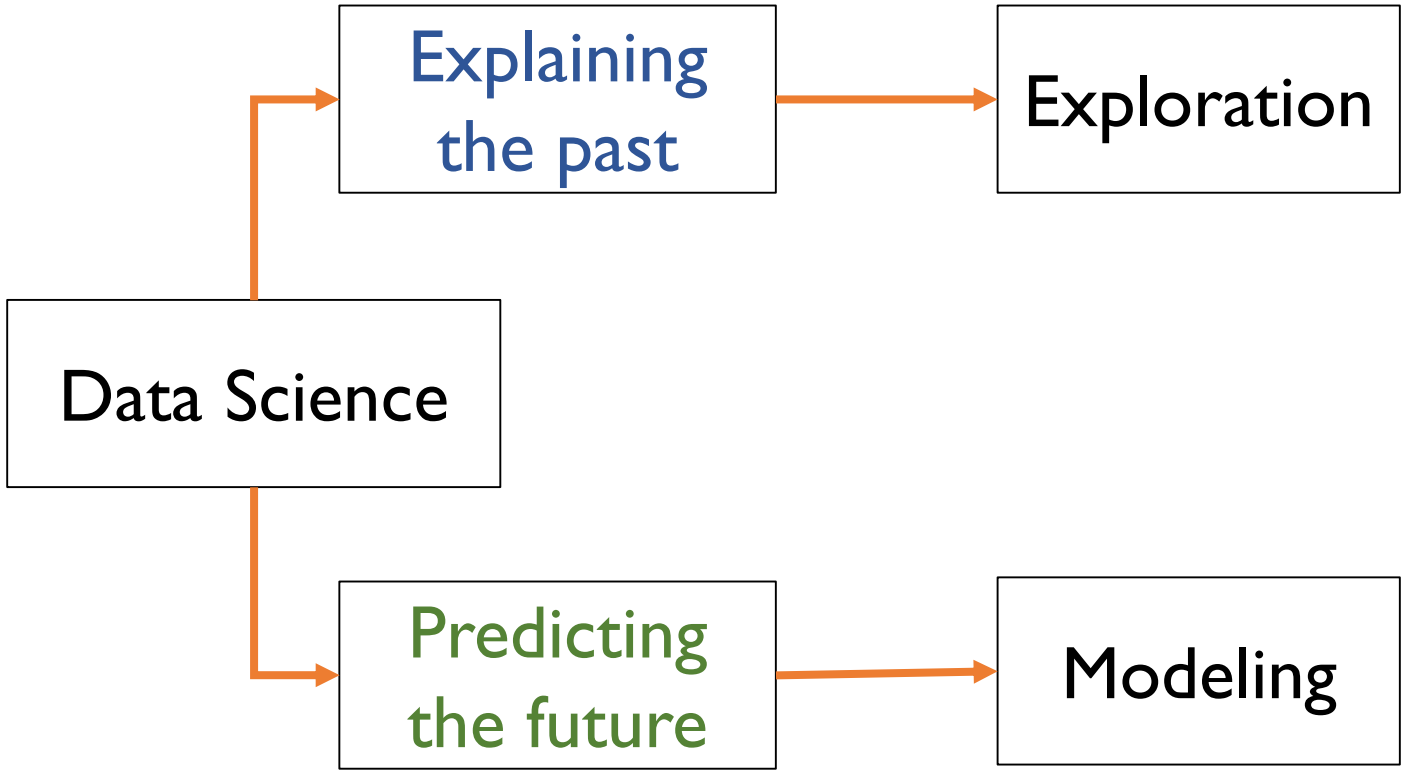
Example Dataset: Fitness Tracker Data



Exploratory Q1: What are the most common reasons for users to switch to a competitor's fitness tracking app?

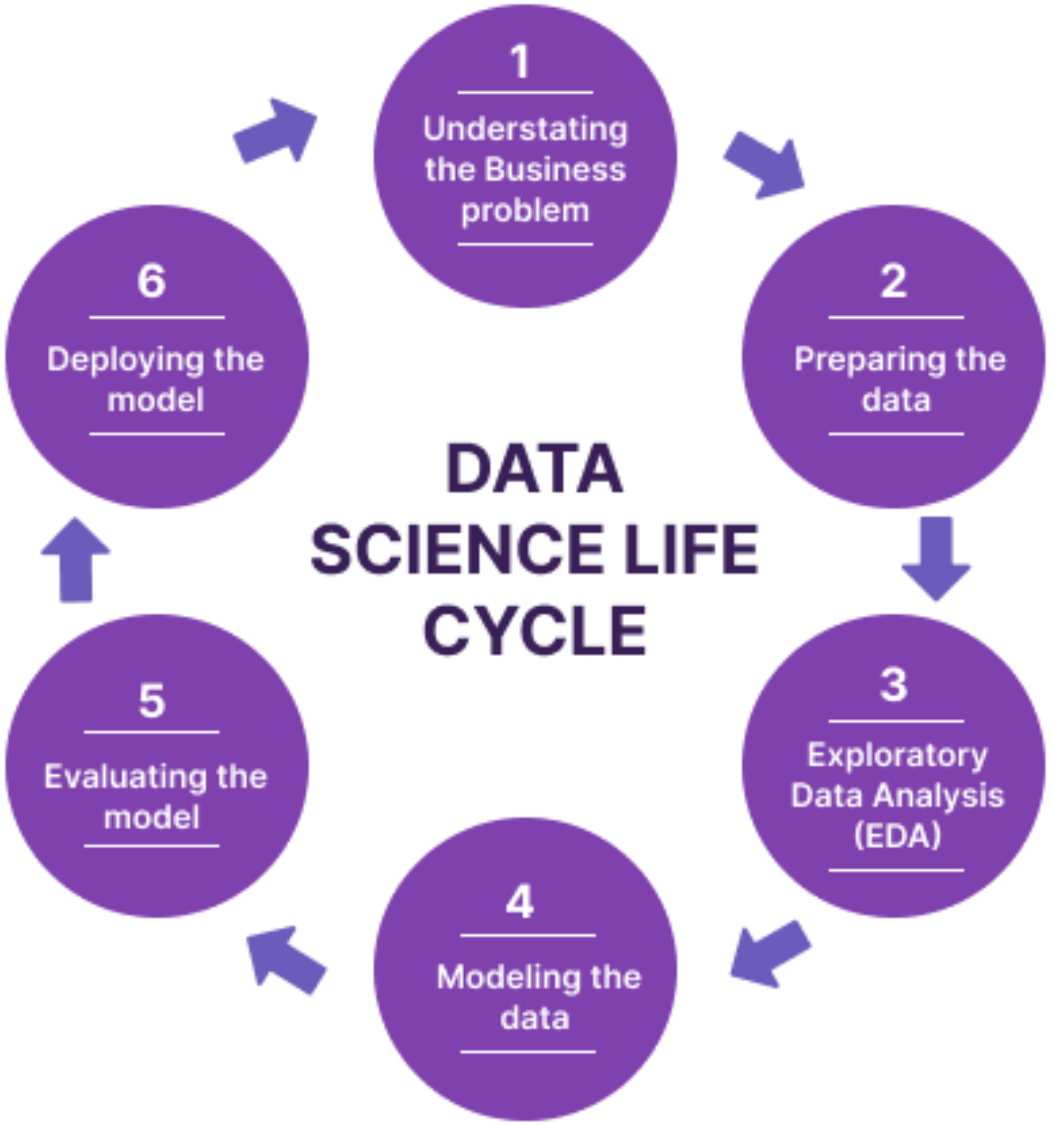
Prediction Q1: Which users are most likely to switch to a competitor's fitness tracking app?

Example Dataset: Fitness Tracker Data



Exploratory Q2: What is the average number of times users open the fitness tracking app per day?

Prediction Q2: What is the ideal time and frequency for sending marketing emails to users?



Source: https://www.odinschool.com/blog/data-science-life-cycle-detailed-explanation-2023

Scenario

Imagine you work for a pizza delivery company, and they want to improve delivery times to increase customer satisfaction.

1- Understanding the problem

- The problem is to analyze delivery data and find ways to reduce delivery times.

- What do we know that could impact delivery times:
  - Weather?
  - Neighborhood?
  - Time of the day?
  - Etc.

2- Preparing the data

- **Activity**: Gather data on pizza deliveries, including order timestamps, delivery addresses, distances, and weather conditions.

- **Challenge**: Deal with missing data, format inconsistencies, and clean the dataset for analysis.

3- Exploring the data

- **Activity**: Visualize delivery time distributions, create heatmaps of delivery locations, and analyze correlations between factors like weather and delivery time.

- **Insight**: Discover that delivery times are longer during bad weather, and certain areas have consistently slower deliveries.

4- Building a model

- **Activity**: Develop a predictive model to estimate delivery times based on factors like distance and weather.

- **Approach**: Use regression techniques or machine learning algorithms to build the model.

- **Experimentation**: Try different models and evaluate their performance.

5- Evaluating a model

- **Activity**: Evaluate the model's accuracy and effectiveness in predicting delivery times.

- **Metrics**: Use metrics like Mean Absolute Error (MAE) or Root Mean Squared Error (RMSE) to assess model performance.

- **Feedback**: Consider customer feedback and practicality in real-world delivery scenarios.

6- Deploying a model

- **Activity**: Deploy the trained model into the pizza delivery system.

- **Integration**: Integrate it with the ordering platform to provide real-time delivery time estimates to customers.

- **Monitoring**: Continuously monitor model performance and retrain it with new data to improve accuracy.

Next…

Introduction to Jupyter Notebook